A comparison of algebraic and complex deformations

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1 Introduction

Algebraic varieties that can also be studied as smooth compact manifolds form a class fascinating objects. Geometers are also interested in the moduli problems they constitute, their families and deformations. In this note, we want to do deformation theory of such objects from two different perspectives: algebraic deformations and complex deformations. As our test ground, we pick elliptic curves over the complex numbers which are also (topologically) complex torus. Indeed, forming the lattice $\langle \tau_1, \tau_2 \rangle \subseteq \mathbf{Z}^2$ on the complex affine line $\mathbf{C}$ such that $\frac{\tau_1}{\tau_2} \notin \mathbf{R}$, the quotient $\mathbf{C}/\langle \tau_1, \tau_2 \rangle$ is an elliptic curve. There is this requirement that $\omega = \frac{\tau_1}{\tau_2}$ such that $Im(\omega) > 0$.

For $\omega_1, \omega_2 \in \mathbf{H}$, they give rise to the same curve iff $\omega_1 = \frac{a+b\omega_2}{c+d\omega_2}$ where $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbf{Z})$ that is $\frac{\mathbf{C}}{\langle 1, \omega_1 \rangle} \simeq \frac{\mathbf{C}}{\langle 1, \omega_2 \rangle}$ (biholomorphic in the complex sense). So, $\mathbf{H}/SL_2(\mathbf{Z})$ is the base space of complex deformations of elliptic curves over the complex numbers. Thus a topological torus has only one differentiable C^∞ manifold structure, but many complex manifold structure, all of which are parameterized by the complex upper half plane modulo $SL_2(\mathbf{Z})$.

2 Elliptic functions

From now on we will denote $\Lambda = \langle 1, \omega \rangle$, and define the Weierstrass function $\wp_\Lambda$ relative to the lattice Λ,

$$\wp_\Lambda := \frac{1}{z^2} + \sum_{\Lambda^*} \left(\frac{1}{(z-\omega)^2} - \frac{1}{\omega^2} \right) \tag{1}$$

which is an elliptic function, having poles of order 2 at the lattice points.

The Eisenstein series of weight $2k$ for the lattice Λ is defined as

$$G_{2k}(\Lambda) = \frac{1}{z^2} + \sum_{\Lambda^*} \frac{1}{\omega^{-2k}} \tag{2}$$

The Laurent series expansion about zero of the function $\wp_\Lambda$ gives us the relation

$$\wp_\Lambda = \frac{1}{z^2} + \sum_{k=1}^{\infty} (2k+1) G_{2k+2}(\Lambda) z^{2k} \tag{3}$$

Let $z \in \mathbf{C} - \Lambda$, we have that

$$\wp_\Lambda^{'}(z)^2 = 4\wp_\Lambda(z)^3 - 60G_4(\Lambda)\wp_\Lambda(z) - 140G_6(\Lambda) \tag{4}$$

Showing this uses the facts that the function

$$f_\Lambda(z) = \wp_\Lambda^{'}(z)^2 - 4\wp_\Lambda(z)^3 + 60G_4(\Lambda)\wp_\Lambda(z) + 140G_6(\Lambda) \tag{5}$$

is identically zero and is holomorphic at $z = 0$. Moreover we see that this function $f_\Lambda(z)$ can be found via comparing the expressions,

$$\wp_\Lambda^{'}(z)^2 = 4z^{-6} - 24G_4(\Lambda) - 80G_6(\Lambda) + ... \tag{6}$$

$$\wp_\Lambda(z)^3 = z^{-6} + 9G_4(\Lambda)z^{-2} + 15G_6(\Lambda) + ... \tag{7}$$

$$\wp_\Lambda = z^{-2} + 3G_4(\Lambda)z^{-2} + ... \tag{8}$$

Now equation (4) looks like the Weierstrass equation for the elliptic curve over fields with characteristic not equal to $2, 3$.So the Weierstrass function deeply links the quantities associated to the lattice Λ and the algebraic equation of the elliptic curve. To simplify the notation, let us write

$$g_3 = g_2(\Lambda) = 140G_6(\Lambda),$$

$$g_2 = g_2(\Lambda) = 60G_4(\Lambda)$$

Here's our set up.We now state three important theorems.[5]
Theorem 1:
The polynomial $f(x) = 4x^3 - g_2x - g_3$ has distinct roots, so its discriminant $\Delta(\Lambda) = g_2^3 - 27g_3^2$ is nonzero.
Proof: (Omitted)

Theorem 2 *Given two complex numbers a_2 and a_3 such that $a_2^3 - 27a_3^2 \neq 0$, there exists complex numbers ω_1 and ω_2 whose ratio is not real, and such that*

$$g_2(\langle \omega_1, \omega_2 \rangle) = a_2$$

and

$$g_3(\langle \omega_1, \omega_2 \rangle) = a_3$$

$g_i(\langle \omega_1, \omega_2 \rangle) = g_i(\Lambda)$ *if Λ is the lattice $\omega_1 \mathbf{Z} + \omega_2 \mathbf{Z}$.* [1]

Proof: We split the proof in three cases: when $a_2 = 0$, when $a_3 = 0$ and $a_2 a_3 \neq 0$.

(i) If $a_2 = 0$, then necessarily $a_3 \neq 0$, otherwise $a_2^3 - 27a_3^2 = 0$. Let $\rho = e^{\frac{2\pi i}{3}}$ and let:

$$\omega_1^6 = \frac{g_3(\langle 1, \rho \rangle)}{a_3}, \omega_2 = \rho \omega_1$$

We claim that g_3 is nonzero. To see this, first observe that $g_2(\langle 1, \rho \rangle) = 0$, since:

$$\frac{1}{60} g_2(\langle 1, \rho \rangle) = \sum_{\omega \in \langle 1, \rho \rangle^*} \omega^{-4} = \sum_{m,n} \frac{1}{(m + n\rho)^4}$$

$$= \sum_{m,n} \frac{1}{(m\rho^3 + n\rho)^4}$$

$$= \frac{1}{\rho^4} \sum_{m,n} \frac{1}{(m\rho^2 + n)^4}$$

$$\frac{1}{\rho} \sum_{m,n} \frac{1}{(n - m - m\rho)^4}$$

since $\rho^2 + \rho + 1 = 0$

$$\frac{1}{\rho} \sum_{M,N} \frac{1}{(N + M\rho)^4} = \frac{1}{60\rho} g_2(\langle 1, \rho \rangle).$$

Now for any lattice L we have $\Delta(L) = g_2^3(L) - 27 g_3^2(L) \neq 0$, by theorem 1, so $g_2(\langle 1, \rho \rangle) = 0$ implies that $g_3(\langle 1, \rho \rangle) \neq 0$.Thus in the above definitions we have $\omega_1 \neq 0$ and so the lattice $\langle \omega_1, \omega_2 \rangle$ is well defined. We have:

$$g_2(\langle \omega_1, \omega_2 \rangle) = g_2(\langle \omega_1, \rho \omega_1 \rangle) = \frac{1}{\omega_1^4} g_2(\langle 1, \rho \rangle) = 0 = a_2$$

$$g_3(\langle \omega_1, \omega_2 \rangle) = g_3(\langle \omega_1, \rho \omega_1 \rangle) = \frac{1}{\omega_1^6} g_3(\langle 1, \rho \rangle) = a_3$$

(ii) If $a_3 = 0$ then $a_2 \neq 0$ and we can take

$$\omega_1^4 = \frac{g_2(\langle 1, i \rangle)}{a_3}, \omega_2 = i\omega_1.$$

We can similarly as above show that $g_3(\langle 1, i\rangle) = 0$, which implies that $g_2(\langle 1, i\rangle) \neq 0$,

$$g_2(\langle \omega_1, \omega_2\rangle) = g_2(\langle \omega_1, i\omega_1\rangle) = \frac{1}{\omega_1^4} g_2(\langle 1, i\rangle) = a_2$$

$$g_3(\langle \omega_1, \omega_2\rangle) = g_3(\langle \omega_1, i\omega_1\rangle) = \frac{1}{\omega_1^6} g_3(\langle 1, i\rangle) = a_3 = 0$$

(iii) Finally, if $a_2 a_3 \neq 0$, we use the function

$$j(\langle \tau_1, \tau_2\rangle) = \frac{g_2^3(\langle \tau_1, \tau_2\rangle)}{g_2^3(\langle \tau_1, \tau_2\rangle) - 27 g_3^2(\langle \tau_1, \tau_2\rangle)}$$

The function J only depends on the ratio $\frac{\tau_1}{\tau_2}$. In fact, J is a modular function of weight 0, that is, it is invariant under the transformations $\tau \mapsto \tau + 1$ and $\tau \mapsto \tau^{-1}$.From this we can show that J is surjective with as its fundamental domain the closure of:

$$R_\Gamma = \{\tau \in \mathbf{C} : Im(\tau) \rangle 0, |\tau| \rangle 1, |\tau + \bar{\tau}| = 1\}$$

. This means we can choose a $\tau \in \mathbf{H}$ such that

$$j(\tau) = \frac{a_2^3}{a_2^3 - 27 a_3^2}$$

Then $J(\langle 1, \tau\rangle) \neq 0$ since $a_2 \neq 0$, which implies that $g_2(\langle 1, \tau\rangle) \neq 0$. So we may pick ω_1, ω_2 such that

$$\omega_1^2 = \frac{a_3}{a_2} . \frac{g_3(\langle 1, \tau\rangle)}{g_2(\langle 1, \tau\rangle)}, \omega_2 = \tau\omega_1$$

Then

$$\frac{g_2(\langle \omega_1, \omega_2\rangle)}{g_3(\langle \omega_1, \omega_2\rangle)} = \frac{\omega_1^{-4} g_2(\langle 1, \tau\rangle)}{\omega_1^{-6} g_3(\langle 1, \tau\rangle)} = \omega_1^2 \frac{g_2(\langle 1, \tau\rangle)}{g_3(\langle 1, \tau\rangle)} = \frac{a_2}{a_3}$$

so that

$$g_3(\langle \omega_1, \omega_2\rangle) = \frac{a_3}{a_2} g_2(\langle \omega_1, \omega_2\rangle).$$

Now observe that:

$$\frac{J(\tau) - 1}{J(\tau)} = \frac{27 a_3^2}{a_2^3} = \frac{27 g_3^2(\langle \omega_1, \omega_2\rangle)}{g_2^3(\langle \omega_1, \omega_2\rangle)}$$

$$= \frac{27 (\frac{a_3}{a_2})^2 g_2^2(\langle \omega_1, \omega_2\rangle)}{g_2^3(\langle \omega_1, \omega_2\rangle)} = \frac{27 a_3^2}{a_2^2 g_2(\langle \omega_1, \omega_2\rangle)}$$

. So, $g_2(\langle \omega_1, \omega_2\rangle) = a_2$ and hence $g_3(\langle \omega_1, \omega_2\rangle) = a_3$.

Theorem 3 *Let E be an elliptic curve defined over $\mathbf{C}$ by*

$$E : y^2 = x^3 - Ax - B$$

Then there exists a lattice $\Lambda \subset \mathbf{C}$, unique upto homothety and a complex analytic isomorphism

$$\phi : \mathbf{C}/\Lambda \to E(\mathbf{C})$$

that is also a group homomorphism.[1]
Proof: Since E is an elliptic curve, the discriminant $4A^3 - 27B^2 \neq 0$. We can then Theorem 2 to find a lattice Λ such that

$$g_2(\Lambda) = 4^{\frac{1}{3}} A$$

and

$$g_3(\Lambda) = B$$

The complex torus $\mathbf{C}/\Lambda$ is then isomorphic to the elliptic curve

$$E^{'} : y^2 = 4x^3 - 4^{\frac{1}{3}} Ax - B$$

which is itself isomorphism to $E(\mathbf{C})$ via the change of the variable $x \mapsto \frac{x}{4^{\frac{1}{3}}}$.

3 Varying the complex structure

Let $\bar{\omega}_1, \bar{\omega}_2 \in \mathbf{H}/SL_2(\mathbf{Z})$, so the curves $\mathbf{C}/\langle 1, \bar{\omega}_1 \rangle$ and $\mathbf{C}/\langle 1, \bar{\omega}_2 \rangle$ are different in the complex analytic sense. Now consider the path $\gamma(t) \in \mathbf{H}$ such that $\gamma(a) = \bar{\omega}_1$ and $\gamma(b) = \bar{\omega}_2$, $\forall t \in [a, b]$.
We want to characterize what happens to the Eisenstein series $G_{2k}(\Lambda)$ as we deform the lattices along this path from $\Lambda_{\bar{\omega}_1} = \langle 1, \bar{\omega}_1 \rangle$ to $\Lambda_{\bar{\omega}_2} = \langle 1, \bar{\omega}_2 \rangle$ on the space $\mathbf{H}/SL_2(\mathbf{Z})$.
Now we follow Theorem 2.
We work with non singular elliptic curves, for which the discriminant should be non zero. This condition is reflected on the requirement of this theorem, that is, given complex numbers $a_2, a_3 \in \mathbf{C}$ such that $a_2^3 - 27a_3^2 \neq 0$, there must exist complex numbers $\tau_1, \tau_2 \in \mathbf{C}$ such that $\frac{\tau_1}{\tau_2} \notin \mathbf{R}$ with the following:

$$g_2(\langle \tau_1, \tau_2 \rangle) = a_2$$

$$g_3(\langle \tau_1, \tau_2 \rangle) = a_3$$

Consider the polynomial $f(x, y) \in \mathbf{C}[x, y]$ having the form

$$f(x, y) = y^2 - x^3 + Ax + B$$

so that we may write

$$E = Spec(\frac{\mathbf{C}[x, y]}{\langle f(x, y) \rangle})$$

and we have the cartesian diagram of deformations

$$\begin{array}{ccc} Spec(\frac{\mathbf{C}[x,y]}{\langle f(x,y)\rangle}) = E & \longrightarrow & \mathcal{E} \\ \downarrow & & \downarrow \psi \\ Spec(\mathbf{C}) & \longrightarrow & Spec(\mathbf{C}[A, B]) \end{array}$$

The fibres $\psi^{-1}(p_1) = \mathcal{E}_{p_1}$, $\psi^{-1}(p_2) = \mathcal{E}_{p_2}$ are deformations of the curve E, for example.
Since we are considering nonsingular curves, it must be that

$$4A^3 - 27B^2 \neq 0$$

So, applying Theorem 2 and 3, there must exist a lattice $\langle 1, \omega \rangle = \Lambda$ for which

$$g_2(\Lambda) = 4^{\frac{1}{3}} A$$

$$g_3 = B$$

Then the complex torus $\mathbf{C}/\Lambda$ is isomorphic to the elliptic curve $Spec(\frac{\mathbf{C}[x,y]}{\langle f(x,y)\rangle})$, where

$$f(x, y) = y^2 - 4x^3 + 4^{\frac{1}{3}} Ax + B \tag{9}$$

This curve is isomorphic to E via the coordinate change $x \mapsto \frac{x}{4^{\frac{1}{3}}}$.

For the two lattices

$$\Lambda_{\bar{\omega}_1} = \langle 1, \bar{\omega}_1 \rangle$$
$$\Lambda_{\bar{\omega}_2} = \langle 1, \bar{\omega}_2 \rangle$$

Then we must have

$$4^{-\frac{1}{3}} g_2(\Lambda_{\bar{\omega}_1}) = A_{\bar{\omega}_1} \tag{10}$$
$$4^{-\frac{1}{3}} g_2(\Lambda_{\bar{\omega}_2}) = A_{\bar{\omega}_2} \tag{11}$$
$$g_3(\Lambda_{\bar{\omega}_1}) = B_{\bar{\omega}_1} \tag{12}$$
$$g_3(\Lambda_{\bar{\omega}_2}) = B_{\bar{\omega}_2} \tag{13}$$

Therefore we should have the polynomial equations

$$f_{\bar{\omega}_1}(x,y) = y^2 - 4x^3 + 4^{\frac{1}{3}} A_{\bar{\omega}_1} x + B_{\bar{\omega}_1} \tag{14}$$

$$f_{\bar{\omega}_2}(x,y) = y^2 - 4x^3 + 4^{\frac{1}{3}} A_{\bar{\omega}_2} x + B_{\bar{\omega}_2} \tag{15}$$

such that

$$Spec(\frac{\mathbf{C}[x,y]}{\langle f_{\bar{\omega}_1}(x,y) \rangle}) \simeq \mathbf{C}/\langle 1, \bar{\omega}_1 \rangle \tag{16}$$

$$Spec(\frac{\mathbf{C}[x,y]}{\langle f_{\bar{\omega}_2}(x,y) \rangle}) \simeq \mathbf{C}/\langle 1, \bar{\omega}_2 \rangle \tag{17}$$

So we have the following cartesian diagram of deformations

$$\begin{array}{ccc} \frac{\mathbf{C}}{\langle 1, \bar{\omega}_{1,2} \rangle} \simeq Spec(\frac{\mathbf{C}[x,y]}{\langle f_{\bar{\omega}_{1,2}}(x,y) \rangle}) & \longrightarrow & \mathcal{E}_{1,2} \\ \downarrow & & \downarrow \psi \\ Spec(\mathbf{C}) & \longrightarrow & "Spec(\mathbf{C}[A_{\bar{\omega}_{1,2}}, B_{\bar{\omega}_{1,2}}])" \end{array}$$

We have written the object at the lower right corner in the quotation marks since its not actually a scheme. It seems that along the path $\gamma : [a,b] \to \mathbf{H}/SL_2(Z)$, not only we are deforming the curves as algebraic varieties but also as complex manifolds as well. I am not sure whether the elliptic curves really deform, because it might happen that the coefficients g_2, g_3 might not actually "change". We need to characterize how the Eisenstein series

$$G_{2k}(\Lambda_{\bar{\omega}_{1,2}}) = \frac{1}{z^2} + \sum_{\Lambda^*_{\bar{\omega}_{1,2}}} \frac{1}{\bar{\omega}_{1,2}^{-2k}}$$

actually change along the path $\gamma : [a,b] \to \mathbf{H}/SL_2(\mathbf{Z})$.

4 L functions

We study a special class of non-singular complex projective varieties: those that admit a (good) reduction modulo p (prime) via their families parameterized by some "base" variety and, in the spirit of Weil, we associate L functions with them and shall attempt to reduce the problem of deformations of these manifolds to the study of these L functions.
Let $q = p^m$ be a prime power and $\mathbf{F}_q$ be a finite field with q-elements. Then Weil defined the zeta function associated to the variety over this field $X_0/\mathbf{F}_q$ as,

$$Z(X_0/\mathbf{F}_q, t) = exp(\sum_{m \geq 1} \frac{N_m}{m} t^m)$$

where $|X_0/\mathbf{F}_q| := N_m$ (number of points of the variety) and conjectured some remarkable properties of this function.
We focus on one of these properties: If $X(\mathbf{C})$ is a non-singular complex projective manifold of which the reduction modulo p is $X_0/\mathbf{F}_q$, then the betti numbers of $X(\mathbf{C})$ can be recovered from $Z(X_0/\mathbf{F}_q, t)$. More precisely, $Z(X_0/\mathbf{F}_q, t)$ can be written as a ratio

$$\frac{P_1(t)P_3(t)...P_{2n-1}(t)}{P_0(t)P_2(t)...P_{2n}(t)}$$

where the P_i's are integral polynomials. Then,

$$degP_i(t) = dim_{\mathbf{C}} H^i(X(\mathbf{C}), \mathbf{C})$$

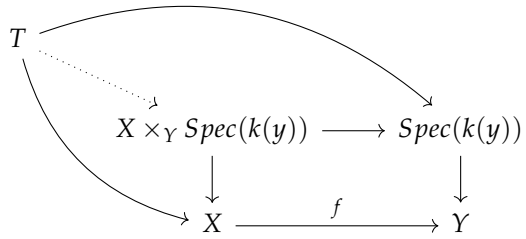
Consider the family $\{X_T(\mathbf{C})\}_{T \in I}$, where each X_T's admit a (good) reduction modulo p, giving us a family $\{X_{0,T}/\mathbf{F}_q\}_{T \in I}$ such that the condition above holds.
We recall that a complex analytic family of compact complex analytic manifolds is given as $\omega : \mathcal{M} \to B$ such that ω is a holomorphic map, for each $T \in B \subset \mathbf{C}^m$, we have that ω^{-1} is compact submanifold of $\mathcal{M}$ and $X_T(\mathbf{C}) = \omega^{-1}(T)$.

As a preliminary definition we know that two (compact) complex manifolds X, Y are called deformations of each other if $\omega^{-1}(T_1) = X$, $\omega^{-1}(T_2) = Y$ for some $T_1, T_2 \in B$. So now we state

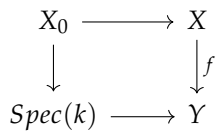
Question : *can we construct a flat morphism of schemes $\omega_p : \mathcal{M}_p \to B'$ such that for some $T_1', T_2' \in B'$ it will hold that $\omega_p^{-1}(T_1') = X_0/\mathbf{F}_q$ and $\omega_p^{-1}(T_2') = X_1/\mathbf{F}_q$ for which the reduction modulo p of $X_0/\mathbf{F}_q$ and $X_1/\mathbf{F}_q$ are precisely X, Y (respectively) that are mentioned above to be deformations of each other?*

Since the betti numbers are topological invariants, the function $Z(X_0/\mathbf{F}_q, t)$ doesn't "care" at all about deforming the complex structure. We are thus motivated to construct a zeta function attached to the reduced modulo p projective non-singular varieties $X_0/\mathbf{F}_q$ (whose complex non-singular version is $X(\mathbf{C})$) that closely depends on the complex structure of $X(\mathbf{C})$.

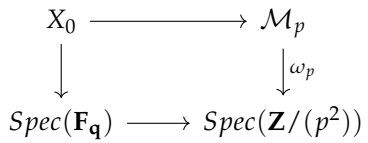
We recall the algebraic definition of deformations (namely those of schemes) (chapter 2 [3]). It is known that the fibre $X_y = f^{-1}(y)$ is topologically the fibre product



where $y \in Y$ is a point and $Spec(k(y))$ is the residue field, for a morphism $f : X \to Y$. We see that X_y is a scheme over $Spec(k(y))$. Take the scheme X_0 over the field k, we define a family of deformations of X_0 to be a morphism $f : X \to Y$ with connected Y together with a point $y_0 \in Y$ such that $k(y_0) = k$ and $X_{y_0} \cong X_0$ and the other fibres X_y are the deformations of X_0. These deformations can be shown by the pullback square as follows



For the scheme X_0 over the finite field $\mathbf{F}_q$ we can take the following example of pullback square as its (arithmetic) deformations



to which we want to compare the diagram

$$\begin{array}{ccc} X(\mathbf{C}) & \longrightarrow & \mathcal{M} \\ \downarrow & & \downarrow \omega \\ \{t_0\} & \longrightarrow & B \end{array}$$

Such that X_0 over $\mathbf{F}_q$ is the reduction modulo p of $X(\mathbf{C})$. Now for a scheme X over $Spec(\mathbf{Z})$ of finite type (whose underlying topological space is $|X|$) the zeta function associated with the scheme is [2]

$$\zeta(X, s) = \prod_{x \in |X|} \frac{1}{1 - (\#k(x))^{-s}}$$

We want to extend this to the Weil zeta functions of the fibres X_y of X_0 over the field k, which are the deformations of X_0.More precisely, keeping Question 1 (from the previous section) in mind, we study schemes X_0 over finite fields $\mathbf{F}_q$ which are reduction modulo p of some complex manifolds $X(\mathbf{C})$.

5 Observations

The moduli problem of (classifying) elliptic curves over $\mathbf{C}$ can be translated to the moduli problem of lattices as the corresponding two moduli functors are isomorphic. Thus, holomorphic deformations of elliptic curves results holomorphic deformations of lattices.
We can construct a unique lattice on $\mathbf{C}$ given an elliptic curve $E(\mathbf{C})$ known as *period lattice*. Let us recall what a lattice in this case is. Given an abelian group Λ of rank 2, we can build a map $\phi : \lambda \to V$ where V is a vector space over $\mathbf{C}$, such that the image $\phi(\Lambda)$ spans V. So our elliptic curve $E(\mathbf{C})$ is the quotient $\frac{V}{\phi(\Lambda)}$.
On the other hand, let us assume that we are given the curve (E, O) with it's marked point O which is a complex torus. Then it must as its first homotopy group have $\pi_1(E; O) \simeq \mathbf{Z} \oplus \mathbf{Z}$. Then the map

$$\phi_* : \pi_1(E; O) \to T_O E$$

can be seen as a lattice. In fact, the tangent space at the marked point O is isomorphic to the dual space of the Dolbeault cohomology group $H^{1,0}_{\bar{\partial}}(E)$. So essectially the lattice is the beautiful map

$$\phi_* : H_1(E; \mathbf{Z}) \to (H^{1,0}_{\bar{\partial}}(E))^*$$

As is known from the theory complex manifolds, the integrable complex structure J induces a deocomposition on the (co)tangent space into holomorphic

and anti holomorphic components, so varying J i.e. varying the complex structure essentially effects the components. As for the complex torus we have the decomposition (of its complex de Rham cohomology)

$$H_{dR}(E, \mathbf{C}) \simeq H^{1,0}_{\bar{\partial}}(E) \oplus H^{0,1}_{\partial}(E)$$

Holomorphically varying (i.e deforming) the complex structure then results varying each of these subspaces (as well as their duals), so the lattice is also holomorphically deformed since $\phi_*(H_1(E;\mathbf{Z})) \subset (H^{1,0}_{\bar{\partial}}(E))^*$.
Now we look at the endomorphism

$$\psi : (H^{1,0}_{\bar{\partial}}(E))^* \to (H^{1,0}_{\bar{\partial}}(E))^*$$

which when restricted to the lattices gives an endomorphism of lattices

$$\psi_l : \phi_*(H_1(E;\mathbf{Z})) \to \phi_*(H_1(E;\mathbf{Z}))$$

So the L function attached to such a period lattice will be **tentatively** defined as

$$\mathrm{L}(\psi_l, t) := \det(1 - t\psi_l)^{-1} = \frac{1}{(1 - a_1 t) \cdots (1 - a_n t)}$$

where a_i's are the eigenvalues of the lattice endomorphism.
Here are some question we were trying to answer:
1) Does holomorphic deformation of elliptic curves results variations of the L functions attached to period lattices?
2) Does varying these L functions (e.g taking derivatives wrt the parameter t) results deformations of the elliptic curves?
To me, the second question seems to be related to the question of deformation of maps between reasonable topological spaces. How do we deform, say, a holomorphic map between two compact complex analytic manifolds? How do we deform a linear map between two vector spaces? Investigating these questions might give us hints to answer the two questions posed above.
If we now assume that the parameter t is a coordinate of the base space of the holomorphic family of lattices, then the above tentative definition of the L function attached to these period lattices doesn't seem to make sense. **How should we modify that?**
Another point of view is worth noting. The square lattice $\mathbf{Z}^2$ can be seen as a scheme $\mathrm{Spec}(\mathbf{Z}[x, y]) \to \mathrm{Spec}\,\mathbf{Z}$. Recently in [4] studied the canonical q-deformation of the de Rham complex, where he also considers the $(q-1)$-adic completion of $\mathbf{Z}$, $\mathbf{Z}[[q-1]]$ (q being a prime power), can be seen as an arithmetic deformation of $\mathbf{Z}$ just as the p-adic completion of $\mathbf{Z}$ i.e. $\mathbf{Z}_p$ can be seen as an arithmetic deformation of $\mathbf{F}_p$. Taking formal spectrum of schemes over the $(q-1)$-adically completed ring $\mathbf{Z}[[q-1]]$ can be seen (or can it?) as family of arithmetic deformations of lattices. There are ways of arithmetically deforming schemes over $\mathrm{Spec}(\mathbf{Z})$ all of which we don't mention here.
The point is, does arithmetically deforming the period lattices of elliptic curves

over **C** that can also be (at least because of *shape*) realized as arithmetic schemes imply complex structural deformations of the elliptic curves themselves.
Of course, complex deformations occur *continuously* where arithmetic deformation is a *discrete* phenomenon.If what we stated above is true, we expect that there exist a discrete set of points in the parameter space of the family of elliptic curves over **C** such that the fibres over those points correspond to the arithemtic deformations of the elliptic curves that are of arithmetic nature.

6 Formal spectrums

Fix a field k. If we are given a scheme $X \to \mathrm{Spec}(k)$, the family schemes of $\mathcal{X} \to \mathrm{Spec}(k[x])/\langle x^2 \rangle$ can be thought of as (in fact they really are) deformations of X. The base space $\mathrm{Spec}(k[x])/\langle x^2 \rangle$ can be thought of as an infinitesimal *thickening* of $\mathrm{Spec}(k)$. Topologically, both of them are just a point, but the functions on the former have derivatives for which we can view them as tangent directions on the underlying topological space of $\mathrm{Spec}(k[x])/\langle x^2 \rangle$ that is just a point. So, as can be guessed easily, they are called first order deformations. The nth order deformations are supposed to be hidden inside $\mathrm{Spec}(k[x])/\langle x^{n+1} \rangle$. But to consider and study deformations of all orders together, we might want to consider spectrums of the ring of formal power series $k[[x]]$. The reason is *inverse limits*.
We start with the notion of inverse limits which are defined for *inverse systems* of algebraic objects (in this we will be considering rings). Consider a collection of rings and ring homomorphisms $(R_n, f_n)_{n \in \mathbf{N}}$ such that they fit into the following sequence

$$.... \to R_{n+1} \overset{f_n}{\to} R_n \overset{f_{n-1}}{\to} R_{n-1} \overset{f_{n-2}}{\to} .... \overset{f_1}{\to} R_1 \overset{f_0}{\to} R_0 \tag{18}$$

We call this collection an inverse system of rings. For such an inverse system the inverse limit is defined as

$$\varprojlim_n R_n := \{\prod_n R_n : f_n(r_{n+1}) = r_n\}$$

Now let us construct a concrete inverse system that looks like the following

$$.... \to \mathbf{Z}/p^{n+1}\mathbf{Z} \overset{f_n}{\to} \mathbf{Z}/p^n\mathbf{Z} \overset{f_{n-1}}{\to} \mathbf{Z}/p^{n-1}\mathbf{Z} \overset{f_{n-2}}{\to} .... \overset{f_1}{\to} \mathbf{Z}/p\mathbf{Z} \overset{f_0}{\to} \{0\} \tag{19}$$

where the ring homomorphism acts as

$$f_n : z + p^{n+1}\mathbf{Z} \mapsto z + p^n\mathbf{Z}$$

so the homomorphism merely decreases the prime powers in the ideals i.e. $\langle p^{n+1} \rangle \mapsto \langle p^n \rangle$
Then taking the inverse limit

$$\varprojlim_n \mathbf{Z}/p^n\mathbf{Z} = \{\prod_n \mathbf{Z}/p^n\mathbf{Z} : f_n(z + \langle p^{n+1} \rangle) = (z + \langle p^n \rangle)\}$$

which is denoted as $\mathbf{Z}_p$, is essentially a ring of finite sequences from each of the sequence from the inverse system and each term in the sequences are related to their adjacent terms via the homomorphisms of the inverse system. These are called p-adic integers. It is curious that on this ring $\mathbf{Z}_p$ a topology can be equipped that is generated by sets of the form

$$z + p^n\mathbf{Z}$$

We say that we have p-adically completed the ring $\mathbf{Z}$ and ended up with a *topological ring* $\mathbf{Z}_p$.
The ring $\mathbf{Z}_p$ is an *arithmetic* deformation of the field $\mathbf{F}_p$. To see that let's ask how we can get to finite fields from the ring of p-adic integers. Basically there exists a short exact sequence

$$0 \to \mathbf{Z}_p \xrightarrow{p^m} \mathbf{Z}_p \xrightarrow{\epsilon_m} \mathbf{Z}/p^m\mathbf{Z} \to 0$$

where the map p^m is multiplication by p^m and ϵ_m is the projection map.
proof of exactness: write...write...write
So for $m = 1$ we get the isomorphism

$$\mathbf{Z}_p/p\mathbf{Z}_p \simeq \mathbf{Z}/p\mathbf{Z} = \mathbf{F}_p$$

To understand the deformation point of view, we need to see how can get from $\mathbf{F}_p$ to $\mathbf{Z}_p$. Thus the ring $\mathbf{Z}_p$ is a ring of mixed characteristic.
This is achieved via the procedure of constructing the ring of Witt vectors $W(k)$ from a given field k. The pieces of the inverse system that was used to construct the ring $\mathbf{Z}_p$ namely the quotient rings $\mathbf{Z}/p^n\mathbf{Z}$ are exactly the "n-th" Witt rings i.e. $W_n(\mathbf{F}_p) = \mathbf{Z}/p^n\mathbf{Z}$. Then the final product is obtained from the inverse limit i.e. $W(\mathbf{F}_p) = \varprojlim_n W_n(\mathbf{F}_p)$.
Recall the construction of crystalline cohomology of a scheme of finite type over a field with positive characteristic. Crystalline cohomologies are just the right derived functor (following Grothendieck's philosophy of how to look at a cohomology theory) of the global section functor of the sheaf on the crystalline site. This site is defined to be the category consisting of objects that are infinitesimal thickenings of open sets of scheme over Witt rings, with divided power structure. I.e objects in this category look like

$$\begin{array}{ccc} U & \longrightarrow & V \\ \downarrow & & \downarrow \\ \mathrm{Spec}(k) & \longrightarrow & \mathrm{Spec}(W(k)) \end{array}$$

Or for the n-th level

$$\begin{array}{ccc} U & \longrightarrow & V_n \\ \downarrow & & \downarrow \\ \mathrm{Spec}(k) & \longrightarrow & \mathrm{Spec}(W_n(k)) \end{array}$$

This clearly tells us why schemes over the Witt rings of finite fields are arithmetic deformations. That's why we may say that the field $\mathbf{F}_p$ itself has been deformed.
We note that as an example of extreme importance, the ring of formal power series over a variable x with coefficients in the field k, $k[[x]]$ is obtained from taking the inverse limit $\varprojlim k[x]/\langle x^n \rangle$. Then taking its usual spectrum $\mathrm{Spec}(k[[x]])$ yields only two points corresponding to the ideals $\langle 0 \rangle, \langle x \rangle$. That's why formal spectrums were constructed to make *geometric* sense of these algebraic deformations.Formal spectrums are ubiquitous for discussing deformations of geometric and algebraic objects in *nicer* ways.
Intuitively, a formal scheme is a scheme that can express the infinitesimal *thickening* of some other scheme where the underlying topological space is the same as that of the original one but the structure sheaf might be different to reflect this thickening (they are essentially sheaves of topological rings which we define below).
We generalize the construction of the ring of p-adic integers via taking inverse limit of an inverse system to the notion of topological rings.
A topological ring R is a ring (equipped with a topology of course) such that the binary operations are continuous mappings with respect to this topology. Like the p-adic ring, we pick an ideal I from the ring R and define I-adic topology. This ideal will be called the *ideal of definition*. So the I-adic topology equipped with the ring R are generated by sets of the form $r + I^n$ where $r \in R$. So in this spirit, we see that for a prime ideal $\langle p \rangle$ being the ideal of definition, the topology equipped with $\mathbf{Z}$ generated by this ideal is $\mathbf{Z}_p$. The topology generated by the ideal $\langle x \rangle$ when equipped with the ring $k[x]$ is the ring $k[[x]]$.
Let us put a further restriction on the topological ring R via imposing that it is Noetherian and let I be the ideal of definition as before. The formal spectrum of such a ring $\mathrm{Spf}(R)$ is defined as the ringed space $(X, \mathcal{O}_X)$ where the topological space X is merely the scheme $\mathrm{Spec}(R/I)$ and the structure sheaf $\mathcal{O}_X$ in any distinguished open set D_f, $\mathcal{O}_X(D_f)$ is the I-adic completion of the ring $R[1/f]$.

7 Functorial machines

Fixing a commutative ring R, we will be considering the category of the local Artin R-algebras that we may denote as **Art**. The deformation functor attached to some scheme X of finite type over fixed ground field $\mathbf{C}$ is

$$\mathrm{Def}_X : \mathbf{Art} \to \mathbf{Set}$$

such that

$$\mathrm{Def}_X(A) := \left\{ \begin{array}{ccc} X & \longrightarrow & X_A \\ \downarrow & & \downarrow \\ \mathrm{Spec}(\mathbf{C}) & \longrightarrow & \mathrm{Spec}(A) \end{array} \right\}$$

(Here we are abusing the notation a bit in the sense that the sets are really the sets of isomorphism classes of these deformation diagrams)
Now let $\tilde{X}$ be the arithmetic object corresponing to the geometric object X i.e. $\tilde{X}$ be the reduction mod p of X for some prime p. Then we attach to $\tilde{X}$ it's functor of arithmetic deformations that we denote $\mathrm{Def}_{p,\tilde{X}}$. We want two groups G, G_p such that the following representations exist

$$\rho : G \to \mathrm{Aut}(\mathrm{Def}_X)$$

$$\rho_p : G_p \to \mathrm{Aut}(\mathrm{Def}_{p,\tilde{X}})$$

Note that by representations of these (no idea what) groups, we mean the attaching automorphisms of the deformations functors to each element of the groups. (That's pretty wild, right??)
Artin L functions are defined for representations of Galois groups. We want to build analogues (or generalizations?) of Artin L functions that are defined for any group representations. And then apply them to the representations of the groups G, G_p (we still don't know how they look) above.
From an (overly?) optimistic view, may be comparing arithmetic and complex structural deformations will reduce to comparing the groups G, G_p. May be there are objects in algebraic geometry for which these two groups are isomorphic, that isomorphism might give us hints of the existence of a new hidden bridge between arithmetic and complex analysis.
Results in this spirit do exist, like the Fontain-Wintenberger theorem that roughly connects characteristic 0 fields with those of positive characteristic. Namely it compares the fields $\mathbf{Q}_p$ and $\mathbf{F}_p[[t]]$. They are by no means isomorphic. But when we extend the former field to $\bigcup_n \mathbf{Q}_p(p^{1/p^n}) = \mathbf{Q}_p(p^{1/p^\infty})$ p-adically complete it and extend the latter to $\bigcup_n \mathbf{F}_p[[t]](t^{1/p^n}) = \mathbf{F}_p[[t]](t^{1/p^\infty})$ followed by t-adically completing it, the theorem states that the absolute Galois groups are isomorphic i.e.

$$G_{\hat{\mathbf{Q}}_p(p^{1/p^\infty})} \simeq G_{\hat{\mathbf{F}}_p[[t]](t^{1/p^\infty})}$$

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